

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 3.2: Probability I — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.2: Probability I
Driving Question	DRIVING QUESTION: If past outcomes of a chance event — like a coin toss or a football match — do not change future probabilities, how can we use mathematics to predict what is likely to happen, and why does this understanding protect us from the dangers of gambling?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Equally Likely Events: Is Gor Mahia "Due" for a Loss?	Learners tossed a 10-shilling coin 40 times per pair, recorded outcomes in a tally table, and pooled class results to compare experimental probability with the theoretical value of $P(\text{Heads}) = \frac{1}{2}$.	Each coin toss is an independent equally likely event. Theoretical probability = $(\text{favourable outcomes}) \div (\text{total equally likely outcomes})$. Past outcomes do NOT change future probability — a coin has no memory — directly contradicting the gambler's fallacy promoted in betting advertisements seen on Nairobi matatus and social media.	The Gor Mahia fan's claim that the team is 'due' for a loss is an example of the gambler's fallacy. Mathematics shows that if each event is independent and equally likely, its probability remains fixed regardless of previous outcomes. Our coin-toss data supported the theoretical $P = 0.5$, especially when the whole class's results were pooled — showing that larger samples give more reliable estimates of true probability.
Lesson 2 Computing Probability of Equally Likely Outcomes	Learners rolled a six-faced die 60 times per group and drew tokens from a bag of githeri-coloured tokens (white/maize, brown/beans, green/peas), recording all outcomes in tally tables and computing experimental probabilities as fractions, decimals, and percentages.	Learners derived the theoretical probability formula $P(\text{event}) = \text{number of favourable outcomes} \div \text{total number of equally likely outcomes}$, and applied it to Kenyan contexts: selecting a name from a class of 30 learners, and drawing a coloured token from a mixed bag. They confirmed that all individual outcome probabilities for a fair die sum to 1.	Learners explained, using the probability formula and the concept of equally likely outcomes, why Gor Mahia FC's past winning streak does not change the probability of the next match result — directly challenging the gambler's fallacy promoted in Nairobi betting advertisements.
Lesson 3 Range of Probability: From Impossible to Certain	Learners analysed Kenyan scenario Evidence Cards (maize kernels, fruit bowls, dice, spinners) and the class's accumulated coin-toss data from Lesson 1 to find	The probability of any event must satisfy $0 \leq P(\text{event}) \leq 1$; $P = 0$ means impossible (e.g., snow in Kisumu in July); $P = 1$ means certain (e.g., sunrise over Mombasa); all	Learners used the range of probability to refute a fictional gambling advertisement claiming '120% odds', and wrote a mathematical explanation of why a sports

	probabilities and place them on a number line.	other events fall strictly between 0 and 1, and no winning streak — including Gor Mahia's — can push a future probability outside these bounds.	commentator claiming Gor Mahia's winning streak raises their probability above 1 is mathematically impossible, using the boundary rule $0 \leq P \leq 1$.
Lesson 4 Mutually Exclusive Events — Identification & The Addition Rule	Learners sorted 10 Kenyan-context event-cards into 'mutually exclusive' and 'not mutually exclusive' piles, justifying each placement. They also rolled a die 30 times each and recorded experimental frequencies for Event A (rolling a 2), Event B (rolling a 5), and Event A or B (rolling a 2 or a 5), pooling class data on the board.	Mutually exclusive events are events that cannot occur simultaneously on a single trial — for example, a matatu cannot arrive in Nairobi and Mombasa at the same time; a Gor Mahia match cannot end in both a WIN and a DRAW. For mutually exclusive events, the addition rule states $P(A \text{ or } B) = P(A) + P(B)$. This rule does NOT apply when events can overlap (e.g., drawing a card that is both a King and a Heart). The addition rule helps decode gambling advertisements that imply bettors can simply sum their chances — but only works when the underlying events truly cannot co-occur.	The experimental die-roll data showed $P(\text{rolling a 2 or a 5}) \approx P(\text{rolling a 2}) + P(\text{rolling a 5}) = 1/6 + 1/6 = 2/6 = 1/3$. The Gor Mahia match analysis, using an assumed probability of 1/3 for each result, showed $P(\text{Win or Draw}) = 1/3 + 1/3 = 2/3$ because WIN and DRAW are mutually exclusive for one match. However, a betting advertiser who highlights 2/3 is hiding $P(\text{Lose}) = 1/3$ and exploiting the fact that across many independent matches, the gambler's losses accumulate — consistent with what Lessons 1–3 established about the independence of each match result.
Lesson 5 Probability of Single-Chance Mutually Exclusive Events — The Addition Rule in Action	We drew coloured tokens (representing plain, coconut, and cardamom mandazi) one at a time from a bag of 15, replaced each token after drawing, and recorded 20 draws per group. We saw that no single draw could produce two flavours simultaneously — the outcomes were mutually exclusive. Our experimental probabilities for $P(\text{plain OR coconut})$ were close to, but not exactly equal to, the theoretical value of 11/15.	When two events are mutually exclusive — meaning they cannot both occur on a single trial — the probability of either one occurring is found by adding their individual probabilities: $P(A \text{ or } B) = P(A) + P(B)$. This rule was confirmed across four different contexts: the mandazi bag, a matatu-route spinner, a standard die, and the school-week days. The rule only applies when events truly cannot co-occur; applying it blindly to overlapping events gives an incorrect (inflated) probability.	Betting advertisements on Nairobi matatus and social media often imply that you can freely combine the probabilities of different outcomes to increase your chances of winning. Today's addition rule reveals this is only mathematically valid when the events are genuinely mutually exclusive and when each individual probability is correctly calculated. In a Gor Mahia match, Win, Draw, and Loss are mutually exclusive — but betting companies frequently present combinations of outcomes in ways that blur this condition, making combined probabilities appear more favourable than they truly are. This is one of the mathematical tricks that makes gambling dangerous.
Lesson 6 Independent Events — The Coin Has No Memory	Learners tossed a coin and rolled a die simultaneously for 24 trials, recording all combined outcomes and organising them into tree diagrams and two-way tables. They found approximately 12 equally likely combined outcomes, each occurring roughly 1/12 of the time. They also drew numbered cards with replacement and noticed the second draw was unaffected by	When two events are independent — meaning the outcome of one gives no information about the other — the probability of both occurring together is found by multiplying their individual probabilities: $P(A \text{ and } B) = P(A) \times P(B)$. Independence means the first event has no memory and cannot influence the second, regardless of what that first outcome was.	Each Gor Mahia football match is an independent event — just like each coin toss and each die roll. The five-win streak carries zero mathematical information about the probability of winning match six. The multiplication rule quantifies joint probability without any adjustment for past results, proving that the gambler's fallacy promoted in betting advertisements on Nairobi

	the first.		matatus is mathematically false: no streak, however long, changes the probability of the next independent event.
Lesson 7 Probability of Independent Events in Context: The Multiplication Rule	Learners tossed a coin (or used a spinner) twice in 20 trials and recorded WIN–WIN outcomes, finding an experimental probability close to 0.25. They constructed probability tree diagrams for two independent coin tosses and for the Kisumu rain scenario ($P(\text{rain})=0.7$), labelling all four branches and verifying that endpoint probabilities summed to 1.	Two events are independent when the outcome of one does not affect the other. The probability of both independent events occurring is found by multiplying their individual probabilities: $P(A \text{ and } B) = P(A) \times P(B)$. This rule was applied to three Kenyan contexts: rain in Kisumu ($0.7 \times 0.7 = 0.49$), two female learners selected ($0.6 \times 0.6 = 0.36$), and a mandazi vendor selling out twice ($0.8 \times 0.8 = 0.64$). The compound probability of two independent events is always less than either individual probability. A betting stall running 180 spins at $P(\text{win})=1/3$ expects to pay out Ksh 1,800 while collecting Ksh 3,600 — an expected profit of Ksh 1,800 on average over many spins (not a guarantee for any short run) — showing that gambling is structured to disadvantage the player.	Because each match Gor Mahia plays is an independent event, the probability of the team winning two consecutive matches is $P(\text{win}) \times P(\text{win})$ — not a different probability influenced by their previous results. The gambler's fallacy claims that past outcomes change future probabilities, but the multiplication rule proves this is false: each event's probability is fixed, and the compound probability of a streak is simply the product of the fixed individual probabilities. This same mathematics explains why betting stalls always profit in the long run — the game is designed so that the house's expected payout is always less than its expected collection, regardless of short-term streaks.
Lesson 8 Consolidation, Final Explanation & DQB Closure: Why Mathematics — Not Luck — Governs the Odds	Learners worked through four Kenyan-context probability scenarios (coin toss independence, Kisumu dice range, mutually exclusive football match outcomes, and a three-match independent-event multiplication), recorded calculations and written explanations on an Evidence Pack, and conducted a gallery walk critiquing peer mini-posters using structured feedback cards.	Each football match (and every chance event studied) is an independent event — its probability is fixed and cannot be changed by past outcomes. The gambler's fallacy is mathematically false because $P(\text{event})$ for an independent trial does not depend on previous results. Gambling is a losing proposition because the multiplication rule for independent events shows that $P(\text{winning multiple consecutive bets})$ decreases sharply with each additional trial. All probability values lie in the range $0 \leq P(E) \leq 1$, and mutually exclusive events (both teams winning the same match) have a combined probability of zero.	Using all four probability concepts together — equally likely events, range of probability, mutually exclusive events, and independent events — learners constructed a complete written and visual explanation of why the Gor Mahia winning-streak debate and the Nairobi matatu betting advertisements are both examples of the gambler's fallacy. The composite class answer, the mini-posters, the DQB closure, and the written assessment all demonstrate that mathematics — not luck, streaks, or intuition — governs the odds, and that this knowledge is a practical tool for responsible decision-making in real life.

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